

Online Appendix to “What Is AI Exposure?”

Measurement Architecture and Statement-Specific Robustness in Early-Career Employment

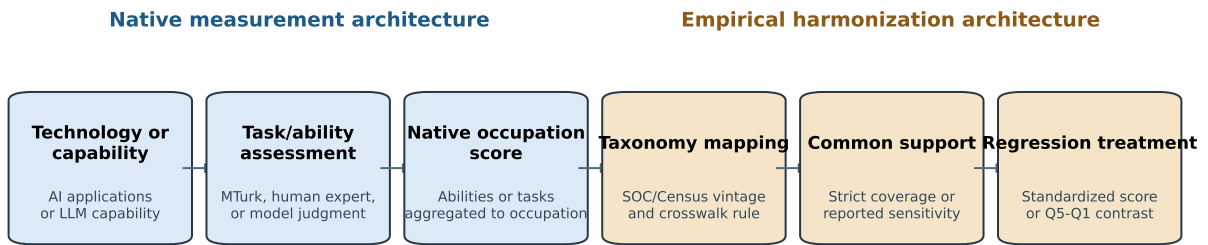
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A. Exposure architectures and measurement lineage

The six architectures differ before any labor outcome enters. Three versions of the ability-based index alter the weighting used to aggregate occupation–ability links. Three task-based measures use the direct-exposure share (alpha), the direct plus complement-software share (beta), and the broad direct-or-complement share. The source files, transformations, and occupation-level checks are recorded in the replication ledger. Figure 1 summarizes the construction paths.



Different choices can change occupational meaning, effective identifying support, or both.

FIGURE 1. Measurement lineage across six AI-exposure architectures. Each path ends in an occupation-level score before merger to the employment data. Ability-based and task-based inputs therefore represent distinct treatment-construction choices, not alternative labels for an observed adoption variable.

TABLE 1. Employment-weighted correlations among exposure architectures

	AIOE-A	AIOE-D	AIOE-S	Alpha	Beta	Broad
<i>Panel A. Pearson correlations</i>						
AIOE administrative	1.000	0.984	0.996	0.295	0.833	0.890
AIOE ability-direct	0.984	1.000	0.981	0.276	0.821	0.883
AIOE source-weighted	0.996	0.981	1.000	0.293	0.832	0.890
Alpha	0.295	0.276	0.293	1.000	0.643	0.346
Beta	0.833	0.821	0.832	0.643	1.000	0.941
Broad	0.890	0.883	0.890	0.346	0.941	1.000
<i>Panel B. Average-rank correlations</i>						
AIOE administrative	1.000	0.987	0.993	0.274	0.844	0.874
AIOE ability-direct	0.987	1.000	0.982	0.258	0.845	0.880
AIOE source-weighted	0.993	0.982	1.000	0.269	0.843	0.874
Alpha	0.274	0.258	0.269	1.000	0.567	0.369
Beta	0.844	0.845	0.843	0.567	1.000	0.961
Broad	0.874	0.880	0.874	0.369	0.961	1.000

Notes: Frozen treatment-side reference sample of 463 occupations, weighted by pre-period employment. Panel B is the weighted Pearson correlation of deterministic average ranks. These are descriptive treatment-side comparisons.

TABLE 2. Employment-weighted correlations with occupational characteristics

Architecture	Cognitive	Manual	RTI	Educ.	Wage	Remote	STEM	Computer
AIOE administrative	.653	-.936	.161	.688	.611	.741	.249	.849
AIOE ability-direct	.689	-.913	.152	.708	.640	.746	.242	.847
AIOE source-weighted	.660	-.939	.148	.700	.617	.754	.249	.854
Alpha	-.032	-.266	.217	-.034	.011	.200	.436	.304
Beta	.478	-.758	.169	.425	.478	.589	.371	.797
Broad	.598	-.810	.108	.530	.600	.618	.253	.833

Notes: Descriptive Pearson correlations using frozen pre-period employment weights. RTI is routine-task intensity. These characteristics are not validation labels for true AI exposure.

B. Occupation mapping and coverage

The employment files and published indices use different occupational taxonomies. The frozen bridge first maps source occupations to Census 2018 codes, preserves one-to-many route mass, and applies the signed coverage rule before any outcome is read. The literal six-measure common support contains 444 occupations and 83.14 percent of employment in the stock comparison. The corresponding native strict-support beta specification contains 468 occupations.

The mapping audit separates four comparisons: original exposure on original support; repaired exposure on the same support; repaired exposure on expanded support; and expanded support excluding computer and mathematical occupations. This decomposition distinguishes score correction from the composition effect of re-admitting occupations. In particular, repairing the vintage mismatch raises computer-and-mathematical employment coverage from 3.33 to 97.7 percent. That repair is necessary for external validity, but it does not by itself identify a causal AI effect.

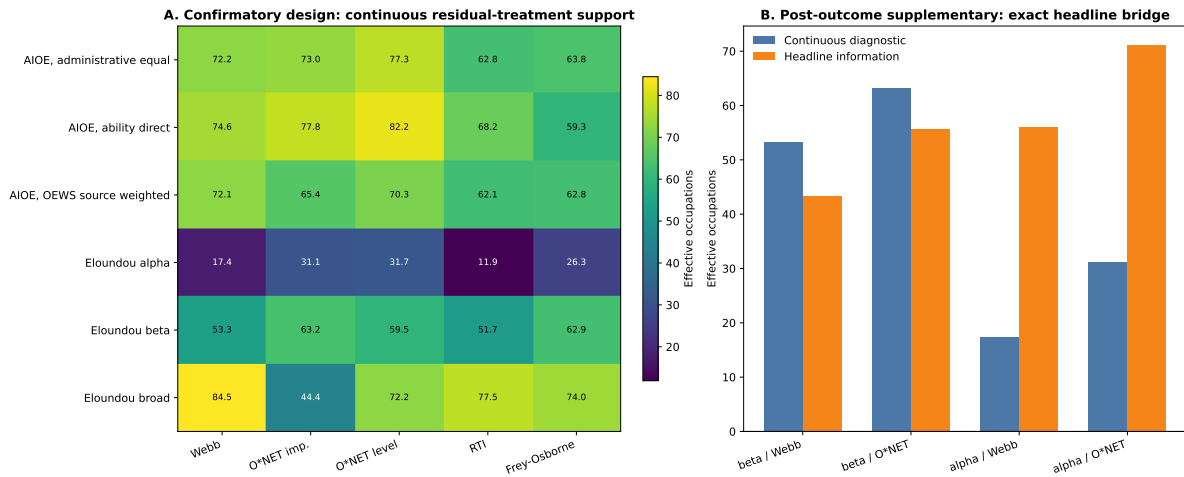


FIGURE 2. Support created by occupation mapping. The figure distinguishes nominal occupational coverage from the residual and estimator-weighted information that supports a particular downstream coefficient.

All crosswalk files are distributed by their public providers or referenced by checksum where redistribution is not appropriate. Route expansion and collapse rules are deterministic. No unmatched occupation is assigned an exposure by occupational title similarity.

C. Residual-treatment and conditional-information support

Support depends on the statement being estimated. A continuous horse race consumes residual score variation after controls; a tail comparison consumes weighted group membership; a ranking statement consumes pairwise order. Figure 2 is therefore not a generic data-coverage diagnostic.

For the primary beta tail contrast, estimator information is concentrated more sharply than the nominal count suggests. The frozen information audit reports an effective occupation count of approximately 43.3 for the fitted Q5-versus-Q1 contrast, compared with 468 nominal occupations. The pre-outcome residual-treatment audit reports an effective count of 53.263. These objects have different weights and should not be equated.

The computerization comparison uses Webb’s software-exposure measure as the frozen primary comparison technology. Archived O*NET computer-use measures, routine-task intensity, and Frey–Osborne automation risk are sensitivities. Remotability is a separate proxy and not a measure of computerization. The comparison-technology results show that coefficient magnitude depends on the conditioning estimand; they do not select a uniquely correct exposure measure.

TABLE 3. Effective residual support across comparison technologies

Architecture	Webb	O*NET importance	O*NET level	RTI	Frey–Osborne
AIOE administrative	72.2	73.0	77.3	62.8	63.8
AIOE ability-direct	74.6	77.8	82.2	68.2	59.3
AIOE source-weighted	72.1	65.4	70.3	62.1	62.8
Alpha	17.4	31.1	31.7	11.9	26.3
Beta	53.3	63.2	59.5	51.7	62.9
Broad	84.5	44.4	72.2	77.5	74.0

Notes: Inverse-Herfindahl counts of employment-weighted residual-exposure variance after projecting each exposure on the named comparison technology. Frozen pre-outcome treatment-side analysis; no labor outcome enters.

Top-five residual-variance shares for the same columns are 18.4, 17.9, 16.6, 20.5, and 20.0 percent for administrative AIOE; 15.9, 16.5, 15.0, 17.8, and 20.6 for ability-direct AIOE; 18.1, 19.5, 18.3, 20.6, and 20.2 for source-weighted AIOE; 41.6, 32.8, 33.3, 46.6, and 36.0 for alpha; 22.2, 20.1, 20.6, 22.8, and 19.8 for beta; and 15.8, 25.2, 18.0, 17.3, and 17.1 for broad exposure.

D. Inference and dependence

The stock models use occupation-clustered analytic standard errors and one-step wild-score confidence intervals with stored, common occupation multipliers where paired or

joint statements require covariance preservation. The common-support comparison uses 999 common multipliers. Separate point-estimate signs, the predeclared simultaneous one-sided band, and the intersection–union test are distinct statements.

All six common-support point estimates are negative. The simultaneous one-sided critical value is 2.26036; the administrative-AIOE upper bound is 0.01858, so the predeclared all-six simultaneous-band criterion is not met. The separately auditable intersection–union marginal p -value is 0.045 and does not supersede the declared criterion.

The paired beta-minus-alpha contrast is -0.03240 , with paired standard error 0.03697, 95 percent interval $[-0.10235, 0.03755]$, and $p = 0.403$. The interval supports only the statement that the design does not detect a difference; it does not establish economic equivalence.

An exploratory occupation-by-calendar-month inclusion–exclusion covariance leaves the main inferential pattern unchanged. It is a model-based dependence sensitivity, not a reconstruction of the CPS survey design, because the public analysis extract lacks the strata, PSU, and replicate-weight structure required for that exercise.

E. Employment-stock estimates

The stock outcome is occupation–age-group–month employment, estimated in a conditional Poisson model with saturated occupation–age, occupation–month, and age–month fixed effects. The target is the post-period change in the young-versus-older employment ratio in exposure Q5 relative to Q1, conditional on the frozen comparison-technology interaction. It is neither an individual unemployment probability nor a monotone dose-response parameter.

TABLE 4. Literal common-support stock estimates

Architecture	Coefficient	Analytic SE	Wild CI	p
AIOE administrative	-0.07386	0.04090	$[-0.14915, 0.00143]$	.057
AIOE ability-direct	-0.10285	0.03811	$[-0.17722, -0.02849]$	.005
AIOE source-weighted	-0.10210	0.04223	$[-0.18040, -0.02380]$	.014
Alpha	-0.10132	0.04171	$[-0.18547, -0.01716]$	.018
Beta	-0.12896	0.04517	$[-0.21614, -0.04178]$	.004
Broad	-0.14652	0.04522	$[-0.23285, -0.06018]$	.003

Notes: Exploratory literal common support of 444 occupations, covering 83.14 percent of employment; 108 static-model months. The outcome is grouped employment stock. Each coefficient is the young $\times$ post Q5-versus-Q1 contrast for the named architecture, conditional on the frozen comparison-technology interaction. Inference uses 999 occupation-cluster wild-score draws.

The primary beta/Webb quintile profile relative to Q1 is Q2 -0.08547 , Q3 -0.04779 , Q4 -0.09703 , and Q5 -0.13107 . The non-monotone middle quintiles are why the paper describes an extreme-tail contrast rather than a dose response.

TABLE 5. Complete frozen alpha/beta headline grid

Exposure	Coverage	Comparison technology	Occ.	Coef.	Wild CI
Beta	strict	Webb	468	-.1311	[-.2170, -.0451]
Beta	strict	O*NET importance	465	-.2085	[-.3190, -.0980]
Beta	sibling-imputed	Webb	477	-.1186	[-.2044, -.0328]
Beta	sibling-imputed	O*NET importance	474	-.1744	[-.2831, -.0657]
Beta	renormalized	Webb	480	-.1186	[-.2045, -.0328]
Beta	renormalized	O*NET importance	476	-.1746	[-.2854, -.0638]
Alpha	strict	Webb	468	-.0987	[-.1726, -.0248]
Alpha	strict	O*NET importance	465	-.0971	[-.1842, -.0100]
Alpha	sibling-imputed	Webb	477	-.1087	[-.1873, -.0302]
Alpha	sibling-imputed	O*NET importance	474	-.1002	[-.1869, -.0135]
Alpha	renormalized	Webb	480	-.1087	[-.1856, -.0318]
Alpha	renormalized	O*NET importance	476	-.1002	[-.1827, -.0178]

Notes: Confirmatory frozen grid. Outcome and estimand are the occupation-by-age employment-stock Q5-versus-Q1 young $\times$ post contrast. Coverage labels are natural-language descriptions of the three frozen support rules. Intervals use 999 occupation-cluster wild-score draws.

TABLE 6. Exploratory age profile of the beta tail contrast

Age group	Coefficient	SE	95% interval
18–21	–0.07445	0.06406	[–0.19236, 0.04347]
22–25	–0.09957	0.05295	[–0.20103, 0.00190]
26–30	0.01502	0.03440	[–0.05040, 0.08044]
31–40	0.06449	0.02739	[0.01419, 0.11479]
41–50	0.03441	0.02334	[–0.00911, 0.07793]
51–65	0	—	reference

Notes: Post-outcome exploratory exercise using the frozen beta treatment. Coefficients compare the named age group with ages 51–65. Intervals for the two youngest groups include zero.

F. Dynamics, pretrends, and comparison technologies

The categorical pretrend joint test does not detect a differential Q5-versus-Q1 pretrend in the pre-period. The post path is non-immediate and non-monotone. These diagnostics guard against a simple linear differential trend, but absence of a detected pretrend does not identify generative AI as the cause of a post-2022 break. Plausible contemporaneous

changes include the technology-sector correction, interest rates, return-to-office policy, post-pandemic normalization, and occupational-composition shifts.

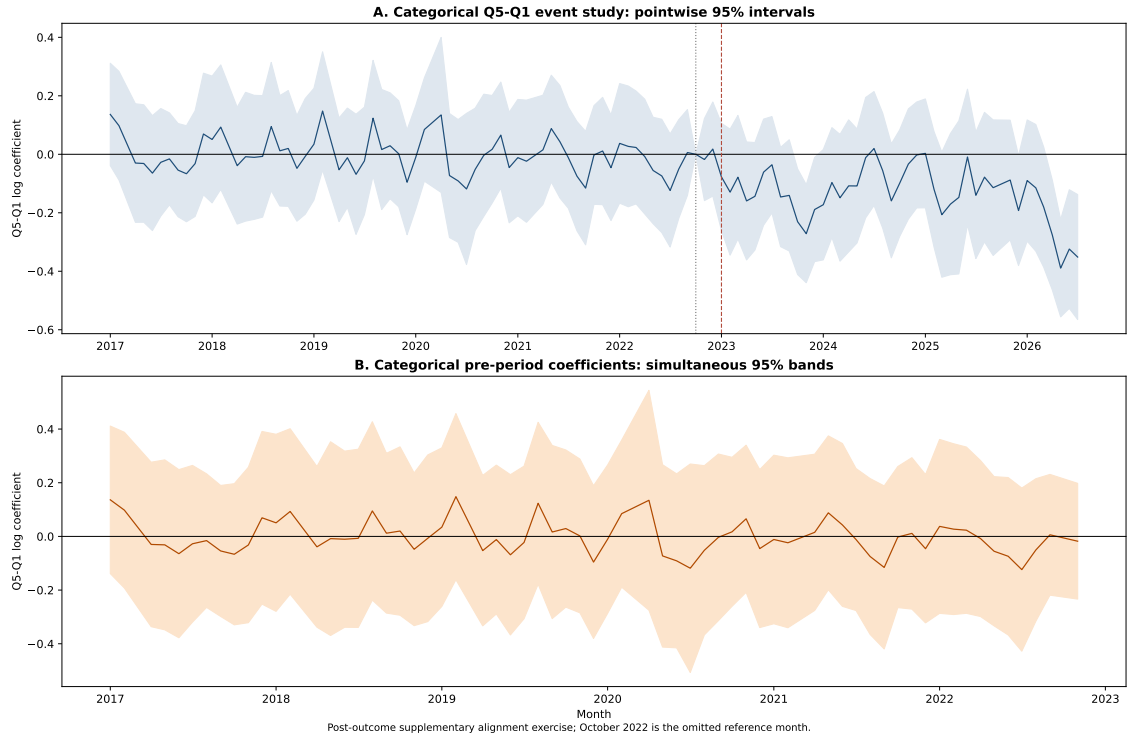


FIGURE 3. Event-time profile for the frozen beta tail contrast. Points and intervals reproduce the sealed exploratory event-study artifact. The figure is a timing diagnostic, not an adoption design.

Remote-work analyses are reported as competing-interpretation sensitivities. They do not absorb the beta gradient, but neither occupation-level feasibility nor realized telework among employed respondents can rule out remote-work mechanisms. In particular, own telework hours are observed conditional on employment and are not used as an individual-level explanation of employment status.

TABLE 7. Comparison-technology and remote-work sensitivities

Specification	AI coefficient	Other coefficient / interval
Beta with Webb	-.13107	—
Beta with O*NET computer-use importance	-.20848	—
Beta with O*NET computer-use level	-.15120	—
Beta with routine-task intensity	-.12771	—
Beta with Frey–Osborne automation risk	-.10011	—
Continuous beta only	-.03814	—
Continuous beta + remote	-.03795	.00469 [–.02288, .03226]
Continuous beta + Webb + remote	-.03718	.00606 [–.02123, .03335]
Remote only	—	-.01884 [–.04508, .00739]

Notes: Existing confirmatory sensitivities. The first five rows are architecture-specific tail estimands; the continuous rows are not numerically commensurate. Remote coefficients are conditional associations, not causal tests of AI against remote work.

The mapping decomposition is $-0.018845 \rightarrow -0.019200 \rightarrow -0.031565$, with -0.029404 after excluding computer and mathematical occupations. The value correction on unchanged support is small; re-admission and composition account for the larger movement. The frozen placebo is 0.001421 with interval $[-0.020395, 0.023236]$ and $p = 0.894$. None of 65 non-reference pre-event intervals excludes zero; six post intervals do. The post-2025 extension-change test has $p = 0.127$.

For the primary model, the occupation-only analytic SE is 0.04441 and the occupation-by-month inclusion-exclusion SE is 0.04494, with normal interval $[-0.21914, -0.04300]$. On common support the corresponding two-way SEs are 0.04106, 0.03810, 0.04213, 0.04127, 0.04578, and 0.04504 for administrative AIOE, ability-direct AIOE, source-weighted AIOE, alpha, beta, and broad exposure.

G. Longitudinal flow estimands and results

All flow models use valid exact adjacent-month CPSIDV links, origin age, official origin LNKFW1MWT, January 2017–November 2022 as pre, January 2023 onward as post, and December 2022 as transition. Links across the MISH 4-to-5 rotation gap are excluded. December 2019 origins are excluded from occupation-dependent transitions.

For exit, the risk set is employed origins and the event is nonemployment at $t + 1$. For occupational outflow, the risk set is employed origins linked to an employed destination and the event is a different harmonized occupation. For entry destination, the risk set is nonemployed origins linked to employment; the outcome is allocation toward destination Q5 rather than Q1. Consequently, the three coefficients are not additive decompositions of the stock coefficient.

TABLE 8. Longitudinal flow estimates

Margin	Coefficient	Analytic SE	Wild CI	p
Employment exit	0.1195	0.0944	$[-0.0668, 0.3059]$	.229
Occupational outflow	0.0107	0.0631	$[-0.1063, 0.1277]$	.857
Persistent outflow	-0.0246	0.1195	$[-0.2580, 0.2088]$	.850
Entry destination	-0.0888	0.0976	$[-0.2787, 0.1011]$	.383

Notes: Post-outcome exploratory conditional-Poisson rate or allocation models using the frozen beta Q5-versus-Q1 contrast and official longitudinal weights. Inference uses 999 occupation-cluster wild-score draws. Each interval includes zero. Entry destination is an allocation estimand, not an employment-finding probability.

Dependent interviewing does not eliminate occupation coding error. The analysis

reports a 9.865 percent immediate-reversal rate and repeats the switch decomposition among legitimate $A \rightarrow B \rightarrow B$ observations. That sensitivity conditions on observing a third adjacent interview and is not a general correction for coding error.

H. Reallocation agreement and the hard benchmark

Among 108,500 realized occupational switches on six-way support, at least one architecture pair assigns opposite exposure directions to 53.28 percent. Pairwise official-weight Cohen kappa ranges from 0.125 to 0.932; standard unweighted Fleiss kappa is 0.501. Pair-specific support is retained rather than silently projecting every pair onto a smaller universal support.

The hard benchmark stratifies by age group, month, origin broad occupational family, and destination broad occupational family. Within 30,170 strata it preserves official-weight detailed origin and destination marginals while destroying observed detailed pairing. Hamilton expansion uses 200,000 pseudo-units and 999 fixed-seed rematches; 84,192 detailed joint cells and 98.31 percent of switch weight are represented. The benchmark mean is 52.32 percent, leaving only a 0.96 percentage-point realized excess. The persistent-switch gap is 0.88 points. Architecture disagreement is frequent, but actual pairings are not meaningfully unusually conflict-heavy under this benchmark.

TABLE 9. Selected pairwise directional agreement

Architecture 1	Architecture 2	Conflict (%)	Cohen κ
AIOE administrative	AIOE source-weighted	3.42	0.932
AIOE administrative	Alpha	39.88	0.178
AIOE ability-direct	Alpha	42.59	0.125
Alpha	Beta	23.46	0.501
Beta	Broad	11.26	0.746

Notes: Descriptive official-longitudinal-weight statistics among realized switches. Conflict is an opposite-sign change for the pair; ties are reported separately in the archived complete 15-pair table. Kappa categories are decrease, tie, and increase.

Conflict declines from 94.59 to 19.06 percent across quintiles of absolute consensus movement. Because the consensus coordinate is constructed from the same families whose directional signs define conflict, this geometry is an organizing decomposition rather than independent behavioral evidence.

TABLE 10. Complete official-weight pairwise agreement table

Architecture 1	Architecture 2	Raw agree	Conflict	κ
AIOE admin	AIOE ability	.9136	.0864	.8271
AIOE admin	AIOE source	.9658	.0342	.9316
AIOE admin	Alpha	.5808	.3988	.1783
AIOE admin	Beta	.7324	.2656	.4658
AIOE admin	Broad	.7659	.2176	.5394
AIOE ability	AIOE source	.8937	.1063	.7874
AIOE ability	Alpha	.5537	.4259	.1252
AIOE ability	Beta	.7208	.2772	.4426
AIOE ability	Broad	.7681	.2154	.5436
AIOE source	Alpha	.5773	.4022	.1715
AIOE source	Beta	.7450	.2530	.4910
AIOE source	Broad	.7713	.2122	.5500
Alpha	Beta	.7449	.2346	.5009
Alpha	Broad	.6168	.3482	.2602
Beta	Broad	.8709	.1126	.7464

Notes: Descriptive results for 108,500 realized switches on six-way support. Raw agreement includes ties; conflict is an opposite-sign pair. Categories for Cohen kappa are decrease, tie, and increase.

I. Consensus, family disagreement, and exact change of basis

Let A_o be the equally weighted centroid of the three standardized ability-based architectures and E_o the corresponding task-based centroid. Define

$$F_o = \frac{A_o + E_o}{2}, \quad G_o = \frac{A_o - E_o}{2}. \quad (1)$$

F is a family-balanced consensus coordinate and G is a family-disagreement coordinate. Both are statistical constructions, not latent causal factors.

The frozen joint model enters standardized F , standardized G , and Webb jointly as continuous young-by-post slopes on 444 occupations and 108 months. With 999 common occupation multipliers, $\hat{b}_F = -0.04036$ with interval $[-0.06925, -0.01147]$, and $\hat{b}_G = 0.03089$ with interval $[0.00180, 0.05999]$. Their centered covariance is -0.00009825 ; the joint max- $|t|$ p -value is 0.008.

The A/E presentation is an exact algebraic change of basis of this same regression. Per weighted standard deviation, the AIOE centroid coefficient is 0.02493 and the Eloundou centroid coefficient is -0.06148 . It is not a second model. The A-minus-E contrast inherits the stored wild-score inference for G exactly; individual A and E intervals use the serialized covariance matrix and the normal critical value because draw-level F/G shifts were not retained.

Treatment-side reconstruction shows that the family-disagreement coordinate is materially affected by alpha. Occupation deletion leaves the fitted G coefficient's sign stable, so the conditional result is not a single-occupation artifact; however, the meaning of the family coordinate remains sensitive to how the task-based family is constructed.

J. Fixed-treatment occupation influence

The leave-one-occupation-out audit deletes each occupation while retaining frozen treatment assignments and refits the same model. It is an influence diagnostic, not a redefinition of treatment. The primary full-sample estimate is -0.131074 ; deletion estimates range from -0.14238 to -0.11055 , with no sign reversal. The largest absolute movement is 0.02052.

Fast food and counter workers are in frozen Q1; deleting them moves the estimate to -0.110553 (movement $+0.020521$). Customer service representatives are in Q5; deleting them moves the estimate to -0.113112 (movement $+0.017962$). Software developers are in Q5 and rank tenth by absolute influence; deleting them moves the estimate to -0.138787 (movement -0.007713). Thus, the result is not a software-developer artifact, while influential

contributions arise at both exposure tails.

TABLE 11. Ten largest absolute occupation-deletion movements

Rank	Deleted occupation	Delete estimate	Movement
1	Fast food and counter workers	−0.11055	+0.02052
2	Customer service representatives	−0.11311	+0.01796
3	Home health aides	−0.14238	−0.01131
4	Computer occupations, all other	−0.12080	+0.01028
5	Office clerks, general	−0.14109	−0.01002
6	Secretaries and administrative assistants ^a	−0.14089	−0.00982
7	Food preparation workers	−0.13981	−0.00874
8	Laborers and freight/material movers, hand	−0.13946	−0.00839
9	Wholesale and manufacturing sales representatives	−0.13924	−0.00817
10	Software developers	−0.13879	−0.00771

Notes: Post-outcome fixed-treatment influence audit. The complete machine-readable table contains all 468 deletion estimates. Employment-stock-weighted signed-movement percentiles (1st, 5th, median, 95th, 99th) are −0.01002, −0.00817, −0.00001, 0.00377, and 0.01796.

^a Except legal, medical, and executive.

For the family-disagreement coefficient, all 444 deletion estimates remain positive between 0.025128 and 0.035993. The ten largest absolute movements follow deletion of driver/sales workers and truck drivers (−0.005766), bookkeeping/accounting/auditing clerks (+0.005100), computer programmers (+0.004516), couriers and messengers (−0.004237), fast food and counter workers (−0.003463), insurance claims and policy-processing clerks (−0.003221), payroll and timekeeping clerks (+0.002869), home health aides (+0.002618), sales managers (+0.002605), and other computer occupations (−0.002509). Employment-stock-weighted signed-movement percentiles (1st, 5th, median, 95th, and 99th) are −0.005766, −0.001693, −0.000024, 0.002155, and 0.004516.

K. Prospective precision and code audit

The pre-outcome headline simulation implied a standard error of 0.01217; the realized analytic standard error was 0.04441, a ratio of approximately 3.65. For the paired beta-minus-alpha contrast, the prospective standard error was 0.01167 and the realized paired standard error was 0.03697, a ratio of approximately 3.17. The frozen paired design therefore had 80 percent power to detect differences of approximately 0.0327 log points, but the prospective precision calculations are design-history provenance and are not evidence of realized power.

The code-only audit shows that the simulation used the actual exposure distribution, employment stocks, design matrix, weights, residual paths, and grouped estimator. It did not independently allocate equal information to all nominal occupations. Treatment concentration and leverage therefore entered mechanically, although no effective-information target was calibrated separately. Comparing $\sqrt{468/43.3} \approx 3.29$ with the precision ratio is only a scale heuristic: the 43.3 value is fitted estimator information, while the prospective residual-treatment effective count was 53.263. The audit does not support a quantitative decomposition. The classification is that the code does not support excessive nominal-information diffusion as the explanation; the available evidence does not isolate a unique cause.

L. Reproducibility provenance

The design freeze peels to commit 22fbf7924809b7a535e31ae0ab68f5b113ce8078; the confirmatory-results tag peels to b16109482c3bf5ca176f6f08976e120b04769945. Frozen receipts record input hashes, code versions, support membership, multiplier seeds, and output hashes. Confirmatory analyses comprise the signed alpha/beta measurement architectures, stock models, paired-difference inference, and mapping and coverage rules. Common-support six-measure comparisons, estimator-support diagnostics, refined event-time analyses, flows, reallocation, and the F/G joint model are labeled exploratory.

The public-use CPS microdata are obtained through IPUMS CPS under its terms of use and are not redistributed in the repository. Exposure inputs and official crosswalks are either vendored where licenses permit or referenced by provider, version, and checksum. Scripts generate analysis files from documented relative paths; credentials, usernames, hostnames, and absolute machine paths are excluded.

This production pass changes prose, layout, tables, and figures only. It estimates no new empirical specification, changes no treatment definition, and creates no new bootstrap multipliers. Numerical claims are inherited from sealed outputs or exact algebraic transformations of them.